



**COMSATS University Islamabad, Sahiwal
Campus**

Project Proposal

(SCOPE DOCUMENT)

for

< UniLost – AI-Based University Lost & Found System >

Version 1.0

By

Ahsan Ellahi CIIT/SP23-BCS-010/ISB

Supervisor

Mr. Fayez Afzaal

Bachelor of Science in Computer Science (2023-2027)

SCOPE DOCUMENT REVISION HISTORY

No.	Comment	Action

Supervisor Signature Date:

Table of Contents

Abstract	4
Introduction	5
Problem Statement	5
Problem Solution for Proposed System	6
Related System Analysis/Literature Review	6
Advantages/Benefits of Proposed System	7
Scope	8
Modules	8
Module 1: Authentication & User Management	8
Module 2: Lost Item Reporting	8
Module 3: Found Item Reporting	9
Module 4: AI-Based Matching System	9
Module 5: Notifications Module	9
Module 6: Chat & Communication Module	9
Module 7: User Profile Management	10
Module 8: Search and Filter System	10
Module 9: Admin Panel	10
Module 10: Location and Map Integration	10
System Limitations/Constraints	10
Software Process Methodology	11
Tools and Technologies	11
Tools	11
Project Stakeholders and Roles	12
Team Members Individual Tasks/Work Division	13
Data Gathering Approach	13
Concepts	13
Gantt chart	14
Mockups	15
Conclusion	18
References	18
Plagiarism Report	18

Project Category: (Mobile Application Development with Artificial Intelligence)

Abstract

UniLost is a smartphone-based lost and found management system designed specifically for university students to help them recover their misplaced belongings efficiently. The system provides a centralized digital platform where users can report their lost and found items with details such as title, description, category, location, and images. Unlike traditional manual processes that depend on notice boards, WhatsApp groups, or word-of-mouth communication, UniLost automates the identification of potential matches using Artificial Intelligence.

The AI-powered matching engine analyzes text descriptions and images of items posted by users to identify similarities between lost and found items. When a strong match is detected, the system sends real-time notifications to both users, enabling quick communication through an integrated chat module. UniLost aims to increase item recovery rates, reduce search efforts, and create a more organized and dependable lost and found ecosystem within the university environment.

Introduction

Lost and found issues are common within university campuses, where students frequently misplace items such as mobile phones, wallets, ID cards, books, and accessories. Traditionally, the process of recovering lost items relies heavily on informal communication channels like WhatsApp groups, Facebook posts, or handwritten notices on departmental boards. These methods are unreliable, unorganized, difficult to search, and often fail to connect the right owner with the right finder. As universities grow in size and student activity increases, the need for a structured and digitalized system becomes essential.

UniLost is proposed as a modern, smartphone-based application that centralizes the lost and found process for university students. The system allows users to post detailed descriptions and images of lost or found items, making the recovery process faster and more accurate. To enhance efficiency, UniLost incorporates Artificial Intelligence to analyze the similarities between lost and found item descriptions. The AI model identifies probable matches and automatically notifies the concerned users.

This document outlines the scope, purpose, modules, methodologies, tools, and development approach of the UniLost system. By providing a clear overview, the document serves as a roadmap for the project's development in the coming semesters. The purpose of this proposal is to establish the foundation for implementing a reliable and intelligent lost and found management system within the university environment.

Problem Statement

Lost and misplaced items are a recurring problem in universities, where hundreds of students move across classrooms, libraries, cafeterias, and outdoor areas every day. Students frequently lose valuable belongings such as mobile phones, wallets, identity cards, USB drives, and books, which often leads to stress, inconvenience, and wasted time. The current process for recovering these items is not systematic; students rely on WhatsApp groups, social media posts, or verbal announcements, which are highly unorganized and inconsistent.

These informal communication channels make it difficult to track, verify, and match lost items with the ones that are found. There is no central platform that maintains a searchable database of lost and found reports. Even when someone finds an item, they often have no idea whom it belongs to or how to reach the owner. Meanwhile, the actual owner struggles to identify if someone has found their item because there is no notification or alert system.

Another major issue is the lack of automation; matching lost and found items manually is time-consuming and often inaccurate. Students may describe the same item differently, which makes it harder to identify matches through simple text search. Additionally, there is no mechanism that facilitates direct communication between the finder and the owner. Without a structured system, many lost items remain unclaimed.

Problem Solution for Proposed System

The proposed system, UniLost, provides a centralized and intelligent platform to solve the issues associated with traditional lost and found processes within the university. The system enables students to easily report lost or found items by uploading details such as title, description, category, location, and images. This organized structure ensures that all relevant information is stored in a searchable digital database instead of being scattered across multiple informal channels.

To address the challenge of manually matching lost and found items, UniLost integrates Artificial Intelligence to analyze both text descriptions and images. The AI engine identifies similarities between newly posted items and existing reports, significantly increasing the accuracy and speed of matching. When a potential match is found, the system automatically notifies the user, removing the need for constant manual checking.

UniLost also includes a real-time chat feature that facilitates direct communication between the finder and the owner. This allows both parties to share additional details, verify identity, and arrange item handover conveniently. The system ensures that users are only contacted when a relevant match is detected, preventing unnecessary messages.

Furthermore, the app uses Firebase for secure authentication, cloud storage, and real-time operations, ensuring reliable and efficient data management. By combining AI-driven matching with a smooth communication flow, UniLost provides a complete solution that reduces user effort, increases item recovery rates, and creates a more structured lost and found environment within the university community. Overall, the proposed system transforms a previously manual and disconnected process into an efficient, intelligent, and user-friendly solution.

Related System Analysis/Literature Review

Several systems and platforms exist that attempt to help users recover lost items, but they lack essential features required for a university-specific environment. WhatsApp and Facebook groups are commonly used by students to post lost or found items, but these platforms are unorganized and do not support structured searching or automated matching. OLX and similar marketplace applications offer lost and found sections, but they do not provide AI-based matching or a dedicated chat system designed for instant communication within a closed community.

Additionally, some universities rely on physical notice boards or manual record-keeping, which makes the process slow and inaccessible to many students. These existing approaches fail to provide a centralized, intelligent, and user-friendly system that efficiently connects finders with the rightful owners. UniLost aims to overcome these limitations by introducing AI-driven matching, real-time notifications, and a dedicated chat system tailored for university users.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
WhatsApp/Facebook Groups	Unorganized, no search, no AI matching, no verification	UniLost provides structured posting, AI-based matching, and secure user authentication
OLX Lost & Found	No automatic matching, designed for public use not universities	UniLost uses AI + real-time notifications and is exclusive to the campus community
Physical Notice Boards	Manual process, no images, no instant reach, easy to miss	UniLost centralizes data, supports multimedia, and instantly notifies users

Advantages/Benefits of Proposed System

1. AI-Based Matching:

UniLost uses artificial intelligence to automatically compare lost and found item descriptions and images, significantly increasing the chances of identifying accurate matches.

2. Real-Time Notifications:

Users immediately receive alerts when a potential match is detected, eliminating the need to manually check posts or social media groups.

3. Centralized and Organized Platform:

All lost and found posts are stored in a structured database, making it easier for students to search, filter, and browse items efficiently.

4. Integrated Chat System:

The built-in chat feature enables direct communication between the finder and the owner, making the handover process quicker and more secure.

5. User-Friendly Mobile Interface:

With a clean and intuitive UI built in Flutter, students can easily upload items, view matches, and manage their profiles without technical difficulty.

6. Campus-Exclusive System:

Only verified university students can use the app, ensuring safety, authenticity, and trust within the platform.

7. Increased Item Recovery Rate:

By combining AI, notifications, and communication tools, UniLost significantly improves the speed and success rate of returning lost items to their owners.

Scope

The scope of UniLost covers the complete digital transformation of the university's lost and found process through a mobile-based application. The system allows users to report lost and found items by providing detailed information such as title, description, images, category, location, and timestamps. Artificial Intelligence is integrated into the system to analyze similarities between lost and found posts, increasing the accuracy and efficiency of identifying potential matches.

When the system detects a match, it automatically sends real-time notifications to the concerned users, ensuring immediate awareness and quick action. Users can then communicate through the built-in chat module to verify item details and arrange for the return. The project also includes user authentication and profile management features to ensure that only verified university students can access and interact with the system.

The platform provides a clean user interface for browsing, searching, and filtering items based on various criteria such as category, date, and location. It also supports multimedia uploads to offer clearer identification of items. Furthermore, UniLost maintains a secure database with the help of Firebase services, ensuring the reliability and safety of stored data.

However, the project does not include GPS real-time item tracking, hardware-based scanning technologies, or cross-campus integration across multiple universities. It also excludes any financial transactions or reward mechanisms for returning items. The system focuses solely on providing a smart, AI-enhanced, reliable, and user-friendly solution for managing lost and found issues within a single university campus.

Modules

Module 1: Authentication & User Management

This module handles the secure login and signup process using Firebase Authentication. Users will create accounts using their university email to ensure authenticity and restrict access to only campus students. The module stores user information such as name, email, campus, and contact details. It also manages password recovery and session handling to ensure smooth access to the application. Additionally, this module includes the storage of FCM tokens for sending real-time notifications. User profile settings, such as updating personal information and profile pictures, are also managed in this module.

Module 2: Lost Item Reporting

This module enables users to submit detailed information about items they have lost. It includes fields such as item title, description, category, location, date, and images. The module uploads images to cloud storage and stores item details in Firestore. It ensures that the information is structured so that AI algorithms can process it effectively. The module also validates user inputs to

maintain data quality. After submission, the lost item becomes publicly visible for potential matching and browsing.

Module 3: Found Item Reporting

This module allows users to report items they have found on the campus. Similar to the Lost Item module, it captures item details including description, pictures, category, and location. Found items are stored in a separate structured dataset, ready for AI comparison. The module ensures that the reporting process is simple and intuitive, encouraging users to upload items they find. The system also helps finders notify the rightful owner quickly through notifications and chat. This module plays a key role in maintaining a complete repository of found items.

Module 4: AI-Based Matching System

This module is responsible for automatically comparing lost and found item descriptions using Artificial Intelligence. It uses text embeddings and image analysis to identify similarities between reports. When a strong match is detected, the system generates a match score and triggers a notification. The AI engine runs on the backend and operates without user input, making the matching process fully automated. This module significantly reduces the time and effort needed to find lost items. It enhances the accuracy of matching, even when item descriptions differ slightly.

Module 5: Notifications Module

This module is responsible for sending real-time push notifications to users when potential matches are detected. It uses Firebase Cloud Messaging to deliver match alerts, updates, and chat notifications. The module ensures that users are immediately informed when someone posts an item that resembles their lost or found report. It also handles notification preferences and ensures that irrelevant or duplicate alerts are minimized. Through this module, communication between users becomes faster and more efficient.

Module 6: Chat & Communication Module

Once a match is detected, this module enables direct communication between the finder and the owner. It provides a secure chat interface where users can exchange messages and verify details. All chat messages are stored in Firestore for real-time updates. The module supports message timestamps, sender identification, and optional image sharing. This ensures smooth communication and a safe environment for item return. The chat system eliminates the need for sharing personal phone numbers or using external apps.

Module 7: User Profile Management

This module allows users to manage their personal profiles within the UniLost application. Users can view and update their information such as name, profile picture, and contact details. It also provides a history of posted lost and found items, enabling users to track their activity. The module ensures that user data is securely stored and easily accessible. It enhances user experience by providing personalization and account management features.

Module 8: Search and Filter System

This module enables users to efficiently search and filter lost and found items within the system. Users can search items based on keywords such as title or description and apply filters like category, location, and date. This functionality improves usability by allowing users to quickly find relevant items without scrolling through all posts. The module ensures fast and accurate retrieval of data from the database.

Module 9: Admin Panel

This module provides administrative control over the UniLost system. The admin can monitor user activities, manage reported or inappropriate posts, and ensure that the platform is used properly. It allows the removal of fake or duplicate entries and helps maintain data integrity. The admin panel ensures system security, reliability, and smooth operation of the application.

Module 10: Location and Map Integration

This module enables users to share and view locations related to lost and found items. Users can specify the location where the item was lost or found, and can also share their meeting location through the chat system. Integration with map services helps users easily navigate and coordinate item return. This feature enhances convenience and improves the overall user experience.

System Limitations/Constraints

1. AI Matching Accuracy:

Although the system uses AI to match lost and found items, the results may not always be 100% accurate due to differences in user descriptions, lighting in images, or incomplete information.

2. Internet Dependency:

UniLost requires a stable internet connection because it relies on Firebase services, cloud storage, and AI processing. Without internet connectivity, users cannot post items, receive notifications, or use chat features.

3. Platform Dependency:

The mobile application is designed primarily for smartphones, and students without smartphones may not fully benefit from the system.

4. Limited to University Community:

The application is restricted to verified university users only, which means it cannot be used outside the campus environment or for public lost-and-found management.

Software Process Methodology

The proposed system will follow the Agile Software Development Methodology, which allows the project to be developed in small, manageable iterations. This methodology is ideal for the UniLost application, as it supports continuous improvement, frequent testing, and the flexibility to incorporate new features such as AI matching and notifications in later semesters. Agile also encourages constant feedback from users, which helps in refining the interface and functionality over time. By adopting this approach, the project will remain adaptable and capable of evolving throughout the 7th and 8th semesters of development.

Tools and Technologies

Tools

Table 2 Tools and Technologies for Proposed Project

Tools / Software	Version	Rationale
Visual Studio Code	Latest	Code editor for Flutter development
Android Studio	Latest	For Android SDK, emulator, and debugging
Firebase Console	Latest	Backend management including Auth, Firestore, FCM
Figma / Pencil Tool	Latest	Used to design UI mockups
Microsoft Word	2019 / Latest	Documentation (Scope document, reports)
Microsoft PowerPoint	2019 / Latest	Presentations for project demonstrations
Technology	Version	Rationale
Flutter (Dart)	Latest	Cross-platform UI framework for mobile development
Firebase Authentication	Latest	Secure user login and signup
Cloud Firestore	Latest	Real-time NoSQL database for storing posts, chats, users
Firebase Cloud	Latest	Send push notifications for matches and chats

Messaging (FCM)		
Firebase Storage	Latest	Store images of lost and found items
AI Embedding API / Model	Latest	Used to generate text and image embeddings for AI matching
Cloud Functions / Server	Latest	Executes AI matching logic and triggers notifications

Project Stakeholders and Roles

Table 3Project Stakeholders for Proposed Project

Project Sponsor	The project sponsor for UniLost is COMSATS University Islamabad, Sahiwal Campus . The university provides the academic environment, resources, and evaluation platform for the development of this project.
Stakeholder	Student Developer: Ahsan Ellahi (SP23-BCS-010) is the primary developer of the system. The student is responsible for requirement analysis, system design, implementation, testing, documentation, and final deployment of the application. Project Supervisor: Mr. Fayeز Afzaal serves as the project supervisor. His role is to guide the student throughout the project, review progress, provide technical and academic feedback, and ensure that the project meets the required academic standards. Co-Supervisor: Mr. Ali Sher Kashif acts as the co-supervisor and provides additional technical support and guidance where required. He assists in monitoring progress and contributes to improving the system's design and implementation. End Users: The end users of the system are the students of COMSATS University. They will use the UniLost application to report lost or found items, receive

notifications, and communicate with other users through the system.

Final Year Project Committee:

The FYP committee is responsible for evaluating the project during different phases and assessing its technical quality, documentation, and overall contribution.

Team Members Individual Tasks/Work Division

Table 4Team Member Work Division for Proposed Project

Student Name	Registration Number	Responsibilities / Modules
Ahsan Ellahi	SP23-BCS-010	Requirement analysis, UI design, Authentication module, Lost & Found item modules, AI-based matching system, Notification module, Chat module, Firebase integration, testing, documentation, and final deployment

Data Gathering Approach

The data and requirements for the UniLost project will be collected using multiple approaches to ensure accuracy and relevance. Informal interviews will be conducted with university students to understand common lost and found issues on campus. Online questionnaires and surveys will also be used to gather feedback regarding user expectations and preferred features. Additionally, observation of existing informal systems such as WhatsApp and Facebook groups will help identify gaps and limitations. This combined approach ensures that the system requirements are practical and aligned with real user needs.

Concepts

Concept 1: Mobile Application Development (Flutter)

This project involves developing a cross-platform mobile application using Flutter. Flutter allows building a single codebase that runs efficiently on Android devices. Through this concept, the student will learn UI design, state management, and mobile app lifecycle handling. It also provides experience in integrating frontend interfaces with backend services.

Concept 2: Firebase Backend Services

Firebase is used as the backend platform for authentication, database management, cloud storage,

and notifications. This concept covers real-time data handling, secure user authentication, and cloud-based storage solutions. The student will gain hands-on experience in managing scalable backend services without maintaining physical servers.

Concept 3: Artificial Intelligence–Based Matching

AI-based matching is a core concept of the UniLost system, where item descriptions and images are analyzed to identify similarities between lost and found posts. This concept introduces semantic text analysis and image comparison techniques. It helps automate decision-making and improves accuracy compared to manual matching methods.

Concept 4: Push Notification Systems (FCM)

Firebase Cloud Messaging is used to send real-time notifications to users when a match is found or a chat message is received. This concept teaches how event-driven communication works in mobile applications. It also ensures timely user engagement even when the application is running in the background.

Concept 5: Real-Time Chat Systems

The chat module enables direct communication between users through real-time messaging. This concept includes message synchronization, real-time updates, and secure data exchange. It provides practical knowledge of building communication systems using cloud-based databases.

Gantt chart

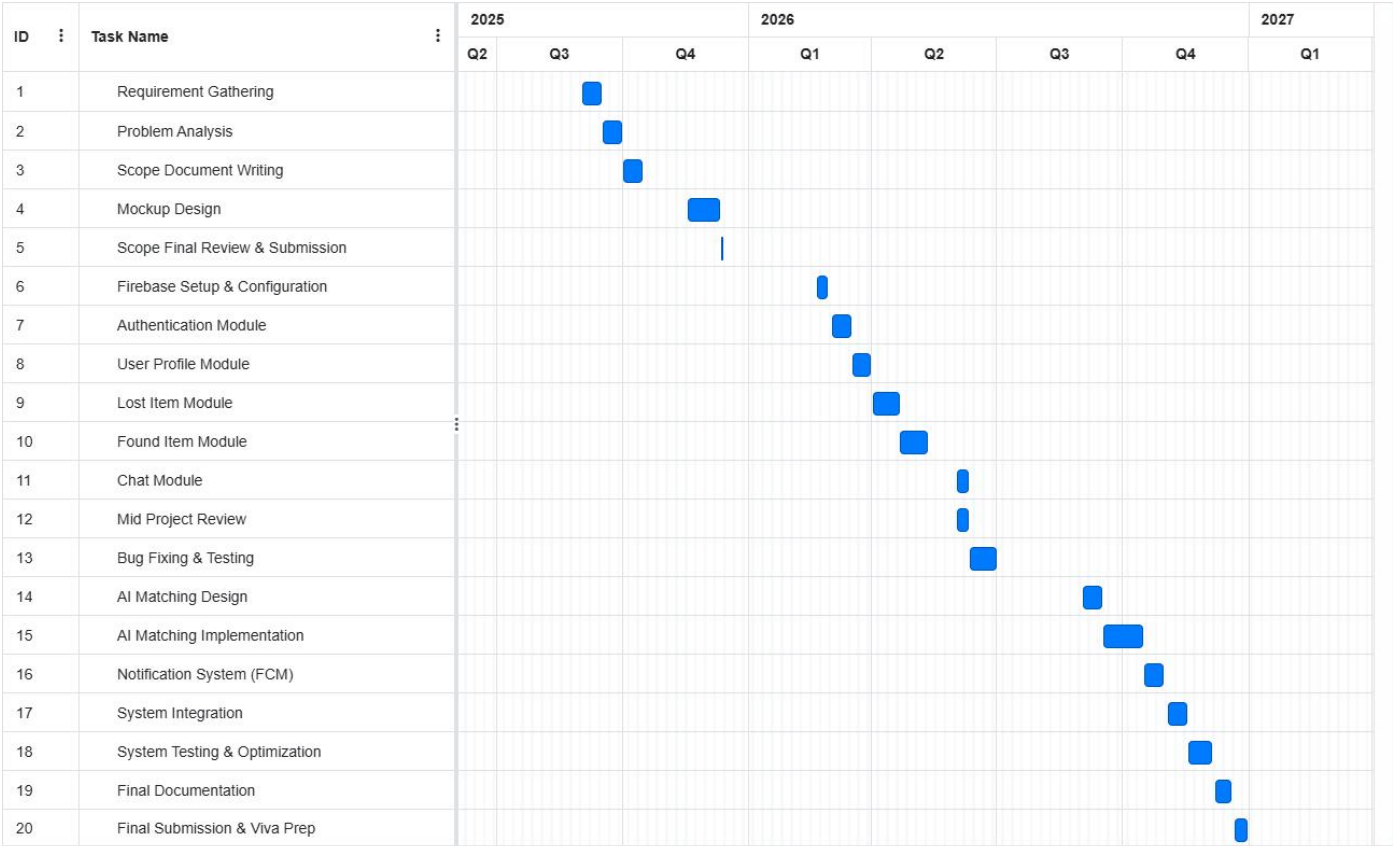


Figure 1Sample Gantt chart

Mockups

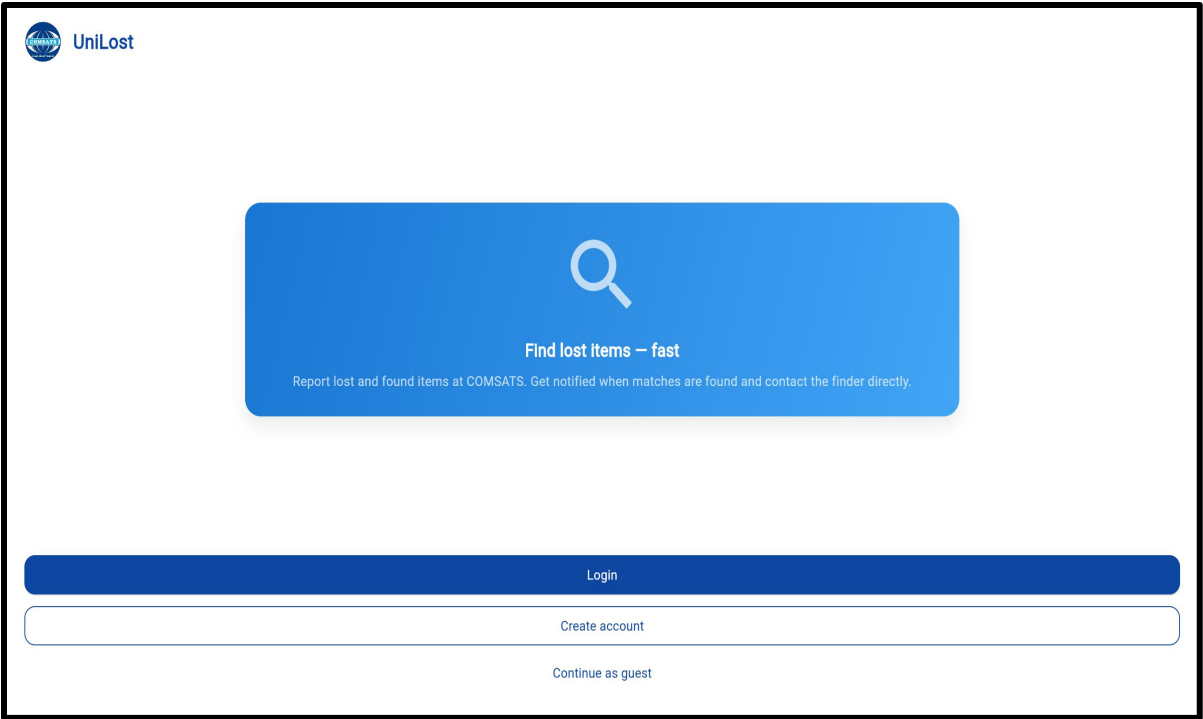
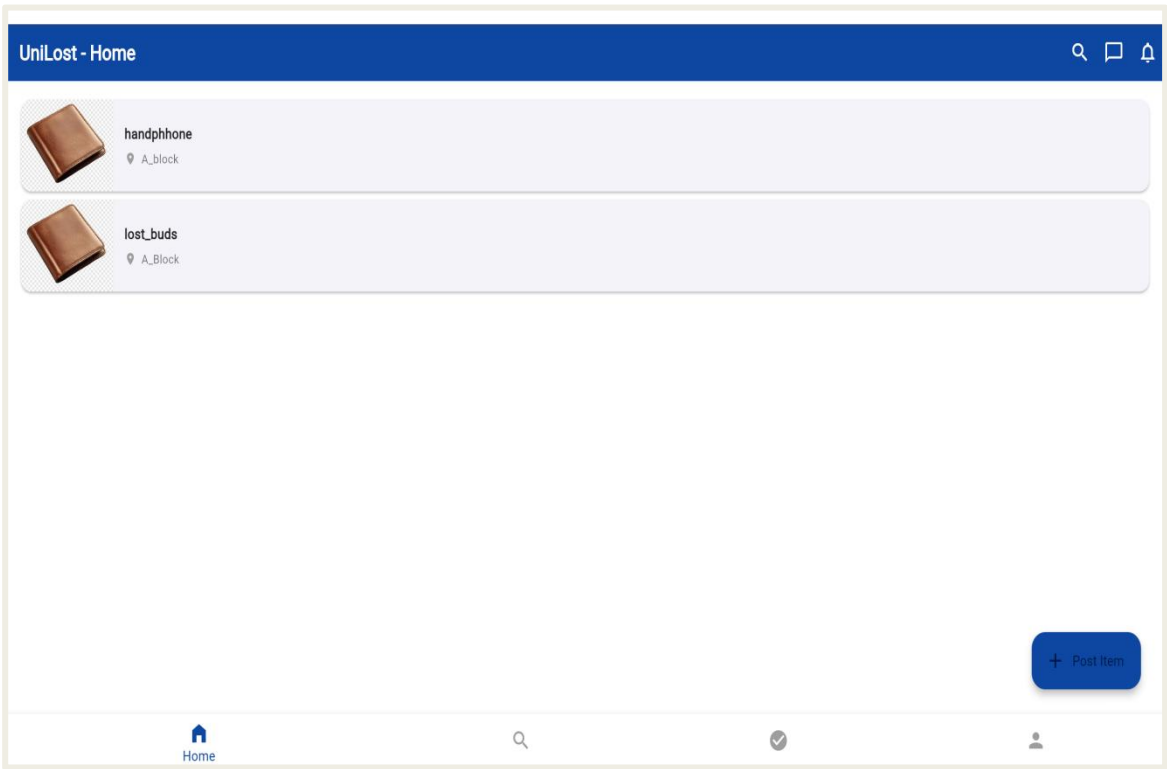


Image 1 Welcome screen :



Report Lost Item

✓ Lost

Found

No image selected

Select Image

Use sample lost

Title

Description

Location (where lost/found)

Submit

Profile



Ali Raza

ali@example.com

Roll: 1000089322

3

Posts

1

Matches

2

Alerts

+92 300 0000000
Phone

ali@example.com
Email

Edit Profile

My Posts

Logout

Settings & Help



Profile

← Settings

Appearance

 Dark mode

Toggle app appearance

Notifications

 Enable notifications

Receive match alerts and messages.

 Auto-match suggestions

Automatically suggest possible matches for your posts.

Account

 Privacy & safety

Control how your data is shared

 Blocked users

Manage blocked users

About

 App info

UniLost

↻ Reset demo settings

← Chat with Owner

Hi, I saw your post about the item.

Type a message...

Conclusion

This scope document presented the proposed design and development plan for **UniLost**, an AI-based lost and found mobile application designed for university students. The project aims to address the common problem of lost and misplaced items on campus by providing a centralized, digital, and user-friendly platform. By integrating features such as item reporting, real-time chat, notifications, and AI-based matching, UniLost offers an efficient alternative to traditional informal methods.

The scope clearly defines the system boundaries, modules, technologies, and development methodology to be followed across the remaining semesters. In the 6th semester, the focus is on documentation and UI mockups, while the 7th and 8th semesters are dedicated to system implementation, AI integration, and final testing. The proposed system is expected to improve item recovery rates, reduce user effort, and enhance communication among students.

Overall, UniLost demonstrates strong potential as a practical and innovative solution to a real-world problem within the university environment. The scope document serves as a foundation for structured development and successful completion of the project in subsequent semesters.

References

- Flutter Documentation. Flutter – Build apps for any screen.
<https://flutter.dev/docs>
- Firebase Documentation. Firebase Authentication, Firestore, Cloud Messaging.
<https://firebase.google.com/docs>
- Google Developers. Firebase Cloud Messaging (FCM) Overview.
<https://developers.google.com/firebase/cloud-messaging>
- OpenAI & AI Research Resources. Text and Image Similarity Concepts.
<https://platform.openai.com/docs>
- COMSATS University Islamabad. Final Year Project Guidelines.
- Pencil Project. Pencil – Interface Prototyping Tool.
<https://pencil.evolus.vn>

Plagiarism Report

The plagiarism check of this project scope document will be performed using the Turnitin tool through COMSATS University Library. The official plagiarism report will be attached after verification by the library staff.